

```
if (row < n) {
```

```
    space--;
```

```
    star += 1;
```

```
    {  
        en {  
            space++;  
            star--;  
        }  
    }
```

→ Cond'n for
assignment question

-	-	-	-	1	-	-	-	-
-	-	-	2	2	2	-	-	-
-	-	3	3	3	3	3	-	-
-	4	4	4	4	4	4	4	-
5	5	5	5	5	5	5	5	5

```
int n = 5, row = 1, star = 1, space = 4, number = 1;
```

```
while (row <= n) {
```

```
    // space
```

```
    int i = 1;
```

```
    while (i <= space) {
```

```
        cout << " ";
```

```
        i++;
    }
```

```
    // star
```

```
    int j = 1;
```

```
    while (j <= star) {
```

```
        cout << number << " ";
```

```
        j++;
```

```
    }
```

```
    cout << endl;
```

```
    number++;
```

```
    row++;
```

```
    space--;
```

```
    star += 2;
```

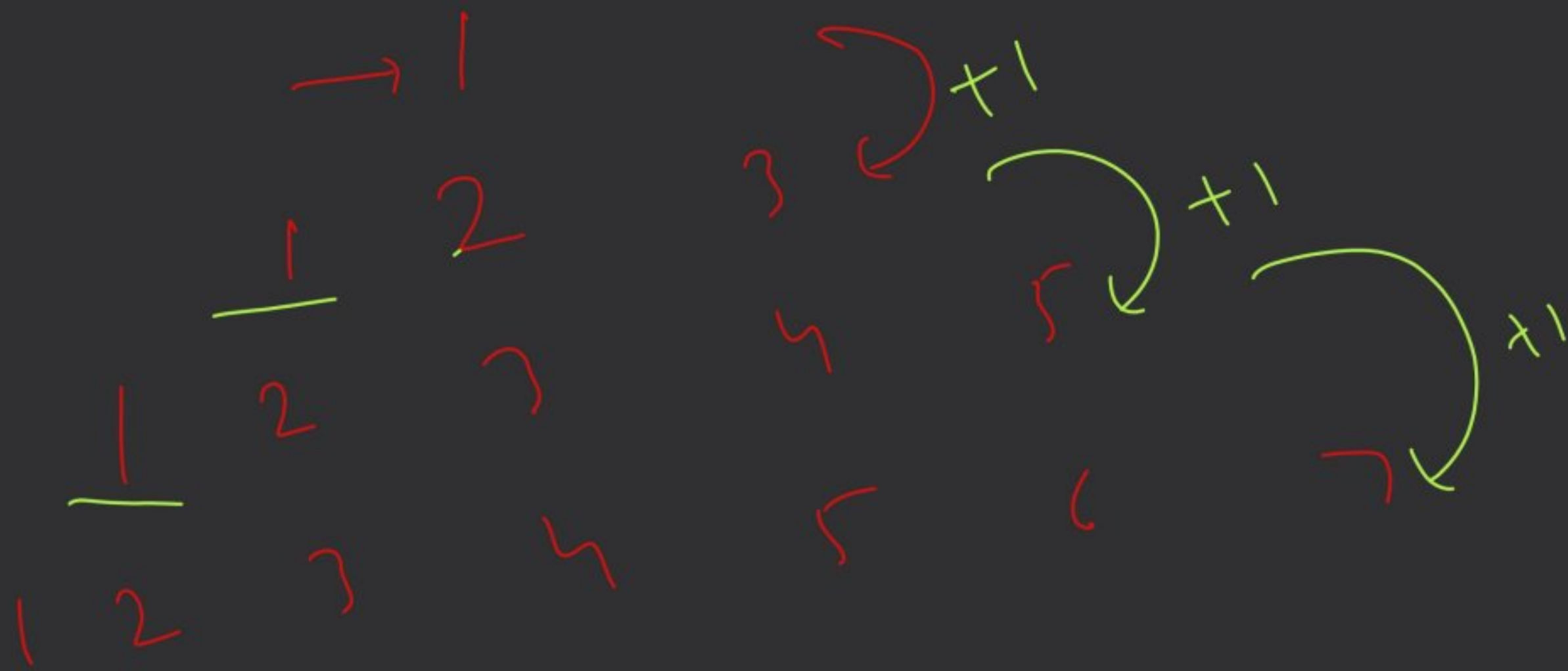
```
}
```

```

      1
    2 2 2
  3 3 3 3 3
```

1
2 3 4
5 6 7 8 9
10 11 12 13 14 15 16

while () {
int number = 1;
}



```
int val = 1;
int j = 1;

while (j <= star) {
    cout << val << " ";
    val++;
    j++;
}
```


W. Prob.



i

n=3

n=5

if (row < n)

$\text{row} < \frac{n}{2} + 1$

* Vertical mirroring only applied ↓

on the numbers. [not on the stars].

~~Imp~~ *

- * Horizontal Mirroring $\rightarrow$ Outer loop
- * Vertical loop $\rightarrow$ Inner loop

$n=5, \text{row}=1, \text{star}=1, \text{space}=n-1;$

```
while (row <= n) {
```

```
int i = 1
```

while (i < space) {

Sont (n, n) ;

$$i + x j$$

3

int j=1;

int val=1;

while (j <= star) {

 Sout (val + " ")

 if (j < $\frac{\text{star}}{2} + 1$) {

 val++;

 } else

 val--;

 j++;

$$\frac{9}{2} + 1 = 4 + 1 = 5 \dots$$

→ 1 2 3 4 5 4 3 2 1

val = ~~8~~ ~~4~~ ~~5~~ ~~4~~ ~~3~~ ~~2~~

j = ~~8~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~

10

↓ ↓ ↓ ↓
1 2 3 4 5

4 3 2 1

← leaves

↑ par1

FileEditSelectionViewGoRunTerminalHelp

EXPLORERLecture_...Pattern_Rombus.classPattern_Rombus.javaPattern1.classPattern1.javaStar_Rhombus.classStar_Rhombus.java

OUTLINETIMELINEJAVA PROJECTS

Pattern1.javaStar_Rhombus.javaPattern_Rombus.java

Star_Rhombus.java

1import java.util.Scanner;
2
3public class Star_Rhombus {
4 public static void main(String[] args) {
5 Scanner sc = new Scanner(System.in);
6 int n = sc.nextInt();
7 int row = 1, space = n-1, star = 1;
8
9 while(row <= 2*n-1) {
10 //space
11 int i = 1;
12 while(i <= space) {
13 System.out.print(s: "\t");
14 i++;
15 }
16
17 //star
18 int j = 1;
19 while(j <= star) {
20 System.out.print(s: "*\t");

Run | Debug

Taking the inputs

Diagram of a rhombus shape with n=5, showing spaces and stars.

cin>>n
alena
input

Ln 4, Col 1 (2 selected)Spaces: 4UTF-8CRLF()JavaGo LivePrettier

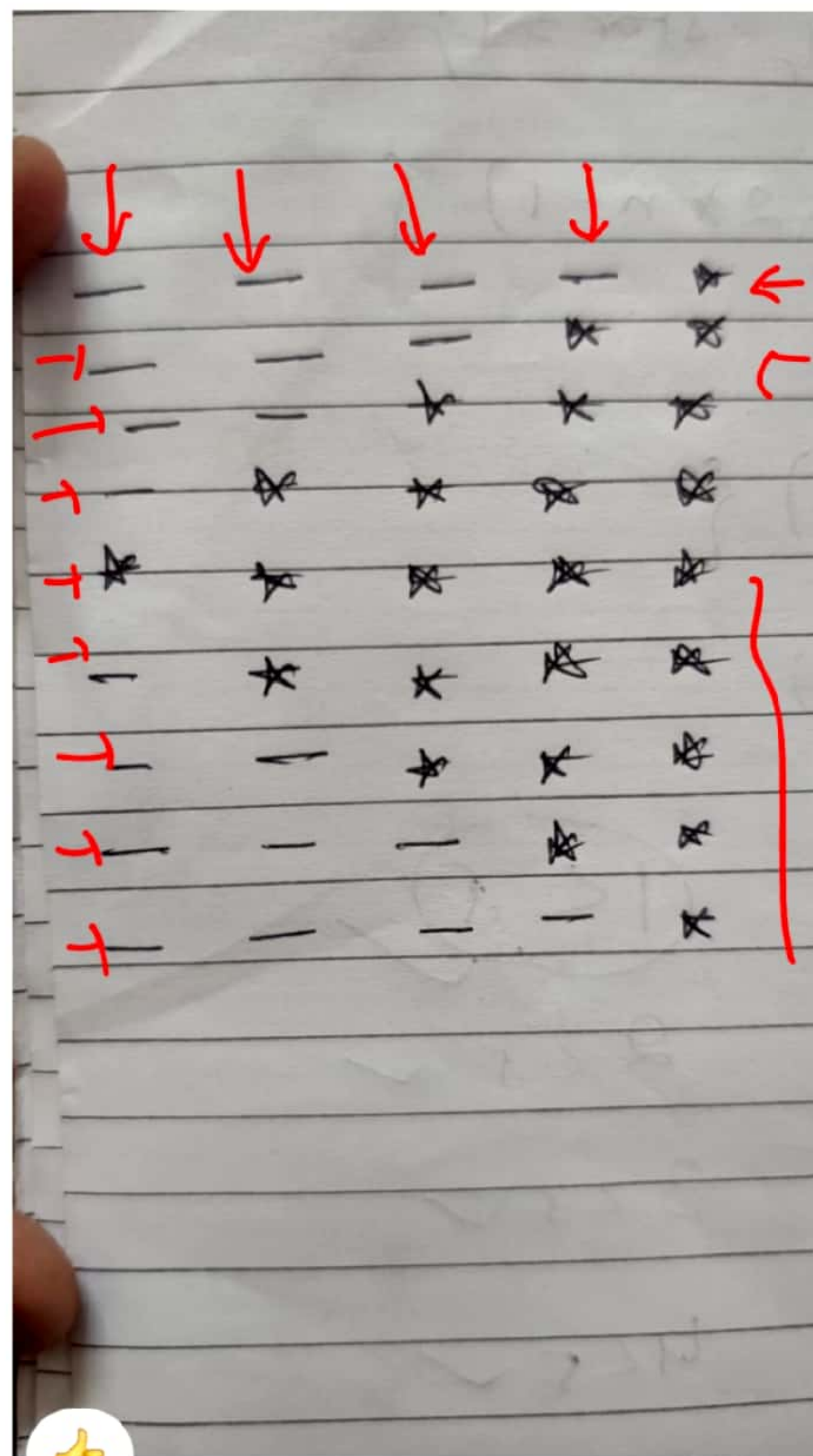
Search

10:37 AM2/26/2026



Gaurav

Yesterday at 7:07 pm



Leaving (space)
Leaving (star)
Leaving (star)
Leaving (space)



@all Do this question on your code editor



FileEditSelectionViewGoRunTerminalHelp

Pattern1.javaPattern2.java XStar_Rhombus.javaPattern_Rombus.java

Pattern2.java > Pattern2 > main(String[])

1 public class Pattern2 {
Run | Debug
2 public static void main(String[] args) {
3 int n = 5, space = n - 1, row = 1;
4 int star = 1;
5
6 while(row <= n) {
7 //space
8 int i = 1;
9 while(i <= space) {
10 System.out.print(s: "\t");
11 i++;
12 }
13
14 //star
15 int j = 1;
16 int val = 1;
17 while(j <= star) {
18 System.out.print(val + "\t");
19 val++;
20 j++;
}

(star = 7)

1 2 3 4 5 6 7

Terminal

1 2 3 4 5 6 7 ←

Ln 11, Col 21Spaces: 4UTF-8CRLF{}JavaGo LivePrettier